Korean Sign Language Recognition under Data Scarcity:
An Ablation Study of Augmentation, Transfer Learning, and Pose-based Modeling

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Abstract

Korean Sign Language (KSL) recognition is under-explored and data-poor: the KSL-77 dataset contains only 1,229 clips over 67 classes ($\sim$*12 per class). Under such* data scarcity*, a standard CNN fails to generalize and overfits to training-signer identity when tested on unseen signers. Using a frozen-VGG16 + LSTM model (LRCN) as baseline, we run a controlled ablation of three strategies: data augmentation, transfer learning, and pose-based modeling. Augmentation* lowers *validation accuracy (*$14.34\% \rightarrow 12.40\%$*), transfer learning from WLASL nearly doubles it to* 23.26%*, and a Pose-LRCN using MediaPipe keypoints is best at* 25.58%*. These results indicate that, under scarcity,* what *the model receives as input (appearance vs. geometry) matters more than* how *it is modeled. We report negative results and the persistent generalization gap honestly, and identify Pose + transfer learning as the most promising untried direction.*

1. Introduction

Sign language serves an estimated 450 million Deaf and hard-of-hearing people [4], and because it differs by region a KSL recognizer cannot be replaced by an ASL or BSL model. Yet KSL recognition remains under-studied and is constrained by the limited scale of public data, which makes generalization, rather than raw model capacity, the central difficulty.

This difficulty is concrete in KSL-77 [6], which contains 1,229 RGB videos over 67 classes from 20 Deaf signers. We adopt a *signer-independent* split, with 16 training and 4 disjoint validation signers, so that the model must learn the sign itself rather than memorize a particular signer; the reported numbers therefore measure true generalization. In this setting, models reach high training but very low validation accuracy, a symptom of *signer overfitting* compounded by *data scarcity* (only $\sim$12 clips per class).

The current state of the art on KSL-77 is Shin *et al.* [4], a CNN+Transformer reaching 89.00%. Rather than chase this number, we ask a more diagnostic question: under a fixed budget, which improvement strategies actually work, and *why*? This framing yields three contributions: (i) a single-variable controlled ablation of augmentation, transfer learning, and pose modeling on a common LRCN backbone; (ii) a counter-intuitive negative result, that augmentation degrades accuracy here, together with an analysis of its cause; and (iii) evidence that pose-keypoint input, which discards appearance, outperforms pixels and sharpens per-class discrimination.

2. Related Work

Isolated word-level sign recognition must jointly model spatial hand shape and temporal motion. The LRCN of Donahue *et al.* [1], which couples per-frame CNN features with a recurrent network, addresses exactly this pairing and serves as our baseline; on KSL, Shin *et al.* [4] push further by combining CNN and Transformer branches to reach 89.00%. Such accuracy is hard to attain when data are scarce, where augmentation and transfer learning are the standard remedies, with transfer typically drawing on a large source corpus such as the word-level ASL dataset WLASL [2] for pre-training. A complementary strategy changes the input rather than the model: to avoid overfitting to appearance, keypoint extractors such as MediaPipe [3] represent only hand and upper-body geometry, an approach that has shown strong results for isolated sign recognition.

3. Method

All models share the signer-independent split, $T{=}32$ frames per clip, and 67 classes; only the per-frame feature extractor and temporal model vary. A clip is $X = \{x_1, \ldots, x_T\}$, $x_t \in \mathbb{R}^{3\times224\times224}$.

3.1. Dataset and Preprocessing

We use KSL-77 [6] (30 fps, 20 Deaf signers); files named `{signer}_{class}.MP4` expose both identities. From each $\sim$4 s clip we sample $T{=}32$ evenly spaced frames via `linspace` (covering start to end; short clips padded by repeating the last frame), resized to 224×224, RGB, and ImageNet-normalized to match the pre-trained VGG16 backbone; frames are cached one folder per clip and stacked into a $(T, 3, 224, 224)$ tensor. Crucially, we split *by signer, not by clip*: 16 vs. 4 disjoint signers (971/258 clips). A clip-level split would leak signer-specific appearance across partitions and inflate accuracy, so our protocol reports strictly on unseen signers.

3.2. Frame Features via VGG16

Per-frame features come from the ImageNet-pretrained VGG16 [5] backbone $\phi(\cdot)$, which stacks 3×3 convolutions and 2×2 max-pooling:

$$a^{(l)} = \mathrm{ReLU}\Big(W^{(l)} * a^{(l-1)} + b^{(l)}\Big), \qquad (1)$$

where $*$ is convolution. We *freeze* the backbone ($\nabla_\phi \mathcal{L}{=}0$) and reuse its generic detectors; pooling and flattening give $f_t = \mathrm{vec}(\phi(x_t)) \in \mathbb{R}^{25088}$.

3.3. Single-Frame CNN Baseline

The baseline discards time and classifies only the central frame, $z = W_2\,\sigma(\mathrm{Drop}(W_1 f_{\mathrm{mid}} + b_1)) + b_2$ ($W_1 \in \mathbb{R}^{512\times 25088}$, $W_2 \in \mathbb{R}^{67\times 512}$, ReLU σ). It cannot separate signs that share a hand shape but differ in motion.

3.4. LRCN: Recurrent Temporal Modeling

LRCN integrates per-frame features with a recurrent network (Fig. 1). Each feature is projected, $e_t = \mathrm{Drop}(\sigma(\mathrm{BN}(W_p f_t + b_p))) \in \mathbb{R}^{512}$, and $\{e_1, \ldots, e_T\}$ is fed to a single-layer LSTM:

$$\begin{aligned}
i_t &= \mathrm{sig}(W_i e_t + U_i h_{t-1} + b_i) \\
f_t^g &= \mathrm{sig}(W_f e_t + U_f h_{t-1} + b_f) \\
o_t &= \mathrm{sig}(W_o e_t + U_o h_{t-1} + b_o) \qquad (2) \\
\tilde{c}_t &= \tanh(W_c e_t + U_c h_{t-1} + b_c) \\
c_t &= f_t^g \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t)
\end{aligned}$$

with input/forget/output gates i_t, f_t^g, o_t, cell state c_t, hidden state $h_t \in \mathbb{R}^{64}$, and elementwise product $\odot$. The forget gate accumulates motion over the whole sign; the clip summary h_T is classified by $z = W_{\mathrm{clf}}\,\mathrm{Drop}(h_T) + b_{\mathrm{clf}}$, $W_{\mathrm{clf}} \in \mathbb{R}^{67\times 64}$. We cap the hidden size at 64 with dropout 0.4, since a larger LSTM memorizes signers and harms generalization. Training uses label-smoothed cross-entropy: with $y_k^{\mathrm{LS}} = (1-\epsilon) y_k + \epsilon/K$ ($K{=}67$, $\epsilon{=}0.1$), $\mathcal{L} = -\sum_k y_k^{\mathrm{LS}} \log p_k$ and $p_k = e^{z_k} / \sum_j e^{z_j}$, which discourages overconfidence.

3.5. Data Augmentation

At training time each frame is randomly transformed, $\tilde{x}_t = \tau(x_t)$ (none at validation). An aggressive setting (v1) collapsed accuracy, so we use a milder v2: horizontal flip ($p{=}0.3$), rotation $\theta \sim \mathcal{U}(-8°, 8°)$, color jitter (0.15), no crop or temporal jitter. Because flips and rotations do not preserve a sign's meaning, even v2 injects label-altering noise (quantified in Sec. 4).

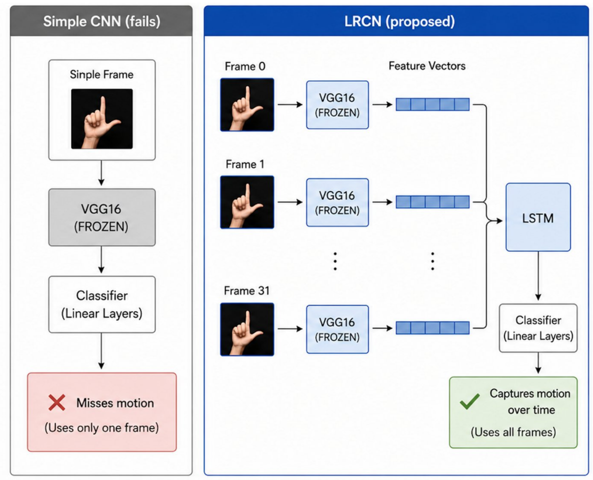

Figure 1. Single-frame CNN (left) vs. LRCN (right). LRCN extracts per-frame features with a frozen VGG16 across 32 frames and models temporal motion with an LSTM.

3.6. Transfer Learning from WLASL

We transfer temporal representations from the word-level ASL dataset WLASL100 [2] (100 classes, 748 clips) in two stages. *(1) Pre-training*: with pre-extracted VGG16 features, we train only the LSTM classifier g_θ, $\theta^* = \arg\min_\theta \mathbb{E}_{(X,y)\sim\mathcal{D}_{\mathrm{WLASL}}}[\mathcal{L}(g_\theta(X), y)]$. *(2) Fine-tuning*: we replace the head ($100 \to 67$), set $\theta_{\mathrm{init}} \leftarrow \theta^*$, and fine-tune on KSL in two phases: a 10-epoch warm-up that updates only the head and LSTM, then 30 epochs that unfreeze the upper VGG layers, with separate learning rates so the pre-trained representation is not destroyed early.

3.7. Combined Configurations

To separate augmentation from transfer, we define Config A (augmentation + transfer) and Config B (transfer only), grid-searching hidden size $\in \{64, 128\}$, dropout $\in \{0.3, 0.4, 0.5\}$, and learning rate $\in \{1\mathrm{e}{-4}, 5\mathrm{e}{-5}\}$.

3.8. Pose-based LRCN

Our proposed model changes *only* the input modality, from pixels to pose keypoints, to test whether the bottleneck is the architecture or the representation (Fig. 2). The temporal module is unchanged; ϕ is replaced by a MediaPipe Holistic [3] extractor giving 53 keypoints per frame (21 per hand, 11 upper-body; lower-body excluded). Each point j has (x_j, y_j, z_j), where depth z_j disambiguates toward/away motion lost in 2D, so

$$f_t^{\mathrm{pose}} = [x_1, y_1, z_1, \ldots, x_{53}, y_{53}, z_{53}] \in \mathbb{R}^{159}, \qquad (3)$$

with missing detections zeroed. For position/scale invariance we center on the nose p_{nose} and normalize by shoulder width $s = \|p_{\mathrm{Lsh}} - p_{\mathrm{Rsh}}\|$: $\hat{p}_j = (p_j - p_{\mathrm{nose}})/s$. The result is

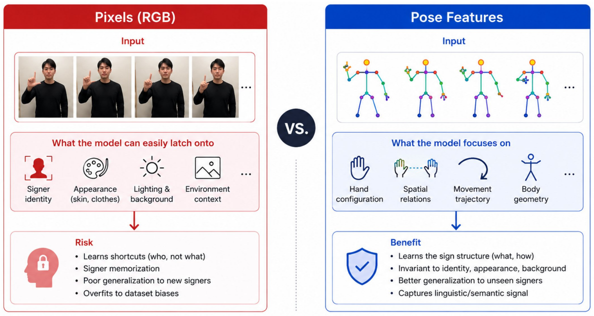

Figure 2. Pixel (RGB) vs. pose input. Pixels risk overfitting to signer identity, appearance, and background; pose emphasizes hand configuration, spatial relations, and motion trajectory, the linguistic signal of the sign.

Model	TL	Aug	Val Acc
Simple CNN (nb02)	–	–	13.57%
LRCN baseline (nb03)	–	–	14.34%
LRCN + Aug (nb04)	–	✓	12.40%
LRCN + WLASL TL (nb05)	✓	–	23.26%
Best Config A (nb06)	✓	✓	21.71%
Best Config B (nb06)	✓	–	25.19%
Pose-LRCN (nb07)	–	–	**25.58%**
Pose-LRCN + Aug (nb08)	–	✓	16.28%
Shin *et al*. 2023 [4]	–	–	89.00%

Table 1. Ablation on KSL-77 (67 classes, signer-split). All models use 32 frames. Shin *et al*. is a 77-class benchmark.

projected $\mathbb{R}^{159} \rightarrow \mathbb{R}^{512}$ and fed to the same LSTM. Because keypoints discard skin tone, lighting, clothing, and background, the model must learn the action's geometry, not the signer's identity. We also define Pose-LRCN+Aug, applying v2 spatial transforms in keypoint space.

4. Experiments and Results

4.1. Setup and Implementation

We report top-1 validation accuracy (971 train, 258 val). Models use PyTorch on one A100 with Adam (weight decay $1\mathrm{e}{-4}$), batch size 8, seed 42; learning rates $1\mathrm{e}{-4}$ (pixel-LRCN) and $1\mathrm{e}{-3}$ (pose-LRCN), with StepLR (step 15, γ=0.5) for WLASL pre-training. Pose-LRCN has only $\sim$0.24M trainable parameters vs. $\sim$13.0M for pixel-LRCN.

4.2. Overall Ablation

Tab. 1 summarizes all models. Temporal modeling adds under 1 pp (CNN 13.57% $\rightarrow$ LRCN 14.34%): with $\sim$12 clips per class the LSTM barely learns motion. Augmentation (12.40%) falls 1.94 pp *below* the baseline. Transfer learning is the largest single gain (+8.92 pp, 23.26%), and pose input is best (25.58%). The gap to the SOTA [4] (89.00%) reflects the structural limits of the data, not the modeling choices alone.

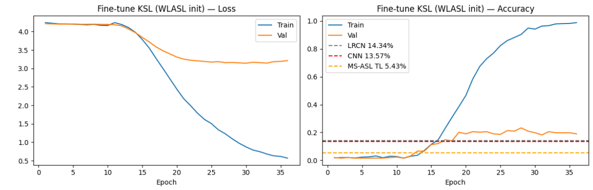

Figure 3. WLASL-initialized KSL fine-tuning. Validation accuracy climbs after $\sim$13–15 epochs, well above the CNN/LRCN baselines.

Run	Config	Hidden	Drop	LR	Val Acc
A1	Aug+TL	64	0.4	1e-4	21.71%
A2	Aug+TL	128	0.4	1e-4	20.16%
A3	Aug+TL	64	0.5	5e-5	18.22%
A4	Aug+TL	128	0.3	1e-4	20.93%
B1	TL only	64	0.4	1e-4	22.48%
B2	TL only	128	0.4	1e-4	24.42%
B3	TL only	64	0.5	5e-5	19.77%
B4	**TL only**	**128**	**0.3**	**1e-4**	**25.19%**

Table 2. Hyperparameter search. Config B (TL only) consistently beats Config A (Aug+TL).

4.3. Why Augmentation Backfires

Augmentation hurt accuracy on LRCN, Config A, and pose alike: aggressive v1 collapsed it to 5.43%, and even mild v2 did not recover the baseline. Since hand direction, position, and speed carry meaning, flips and rotations do not preserve the label, so semantic corruption outweighs regularization. Caveat: v2 also changed dropout and label smoothing, so the effect is not perfectly isolated, and we leave a clean single-variable toggle for future work.

4.4. Why Transfer Learning Helps

In Fig. 3, validation accuracy stays near baseline until $\sim$13–15 epochs, then rises sharply to the best value of any model. This *delayed rise* shows the WLASL-pretrained LSTM is a better starting point than random init; the jump coincides with unfreezing the upper VGG layers during full fine-tuning. Transfer thus improves both final accuracy and the speed of generalization.

4.5. Combined Configurations

In the grid search (Tab. 2), transfer-only Config B beats Config A on every matched pair (e.g. B4 25.19% vs. A4 20.93%); the best, B4 (hidden 128, dropout 0.3), reaches 25.19%. This independently confirms that stacking augmentation on effective transfer lowers accuracy. Fig. 4 contrasts the confusion matrices: the LRCN baseline (14.34%) collapses onto a few columns with almost no diagonal, whereas B4 (25.19%) shows a clear diagonal and resolves many confusions.

4.6. Pose-based Model and Overfitting

Pose-LRCN is best overall at 25.58%. With architecture, hyperparameters, and recipe identical to pixel-LRCN, changing

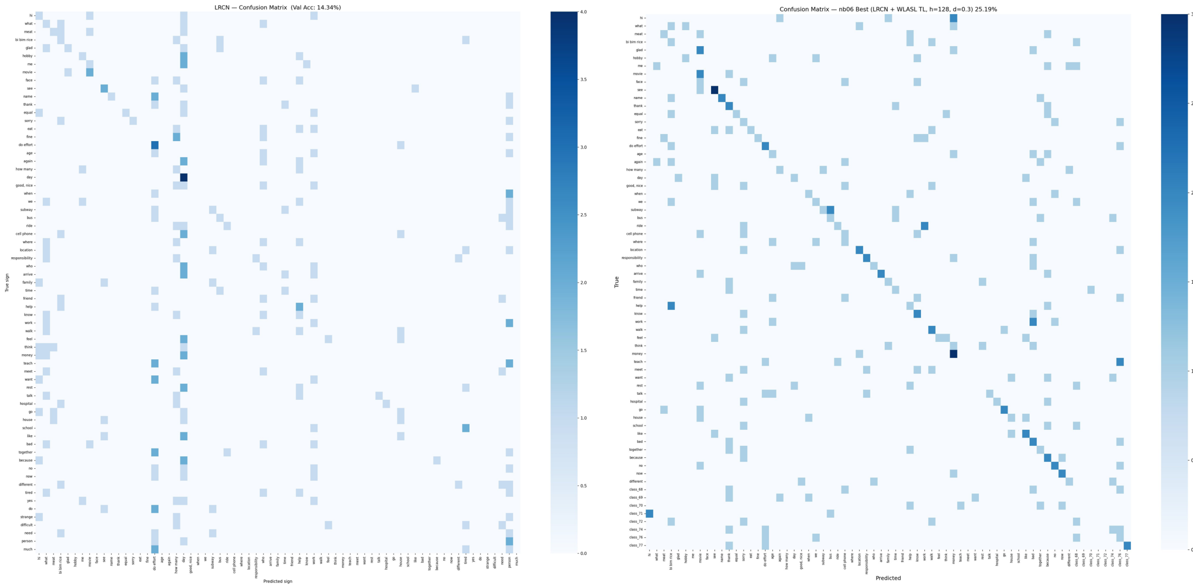

Figure 4. Confusion matrices: LRCN baseline vs. best transfer model. **Left**: LRCN baseline (14.34%, no transfer), where predictions collapse onto a few classes. **Right**: best Config B4 with WLASL transfer (25.19%), where a clear diagonal emerges and many confusions are resolved.

only the input to keypoints improves accuracy by over 11 pp, confirming that the bottleneck is the input representation, not the temporal model; Pose-LRCN+Aug drops to 16.28%, showing augmentation hurts regardless of modality. Every model overfits: LRCN reaches ~84% train vs. 14.34% val, and transfer/pose models exceed 90% train. Standard mitigations (dropout, reduced capacity, label smoothing, early stopping) reduced but did not eliminate it, because the root cause is insufficient *signer diversity*, not model capacity.

5. Discussion and Conclusion

We compared augmentation, transfer learning, and pose-based modeling for KSL recognition under data scarcity, via a single-variable controlled ablation on a common LRCN backbone. Three findings stand out. **(i)** Augmentation backfires: because hand orientation, position, and speed are meaning-bearing, spatial transforms corrupt labels rather than regularize. **(ii)** Transfer learning from WLASL is the most effective single technique, nearly doubling accuracy; its delayed-rise curve indicates a better initialization that generalizes faster. **(iii)** Replacing pixels with pose keypoints gives the best accuracy and the clearest per-class discrimination, by removing the appearance cues that drive signer overfitting.

Limitations and future work. We did not test Pose + transfer learning. Since Pose+Aug hurt but transfer and pose each helped alone, combining the two effective techniques is the most promising untried direction and could plausibly exceed 30% while narrowing the generalization gap. More broadly, our overfitting analysis shows the binding constraint is signer diversity: collecting or synthesizing data from more signers, rather than augmenting existing ones, is the principled next step. The central lesson is that under data scarcity, *what* the model receives as input can matter more than *how* it is modeled.

References

[1] J. Donahue, L. A. Hendricks, S. Guadarrama, M. Rohrbach, S. Venugopalan, K. Saenko, and T. Darrell. Long-term recurrent convolutional networks for visual recognition and description. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, 2015.

[2] D. Li, C. Rodriguez, X. Yu, and H. Li. Word-level deep sign language recognition from video: A new large-scale dataset and methods comparison. In *IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2020.

[3] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, et al. MediaPipe: A framework for building perception pipelines. *arXiv:1906.08172*, 2019.

[4] J. Shin, A. S. M. Miah, M. A. M. Hasan, K. Hirooka, K. Suzuki, H.-S. Lee, and S.-W. Jang. Korean sign language recognition using transformer-based deep neural network. *Applied Sciences*, 13(5):3029, 2023.

[5] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. In *Int. Conf. Learn. Represent. (ICLR)*, 2015.

[6] S. Yang, S. Jung, H. Kang, and C. Kim. Korean sign language dataset for action recognition. In *Int. Conf. Multimedia Modeling (MMM)*, 2020.